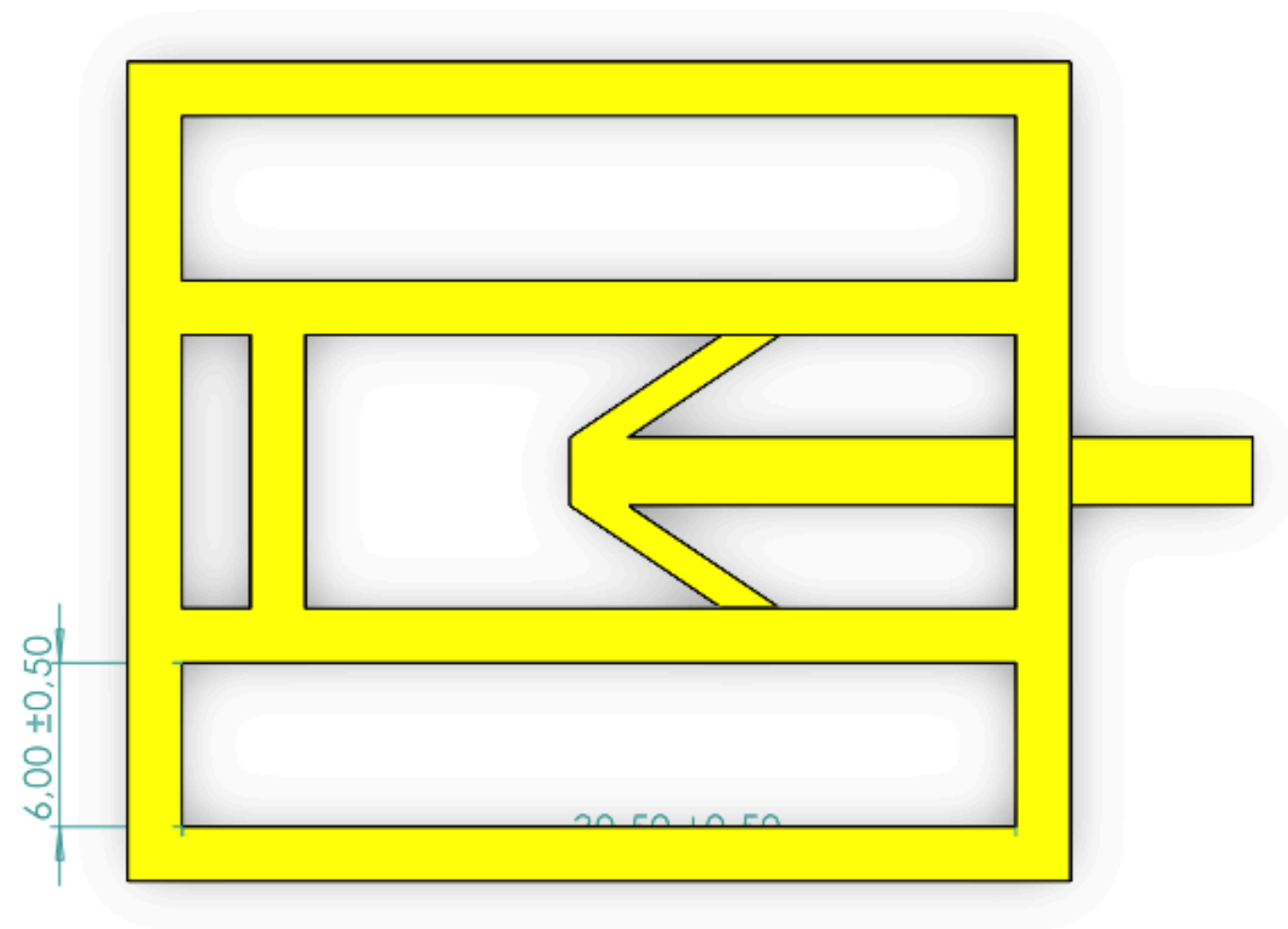


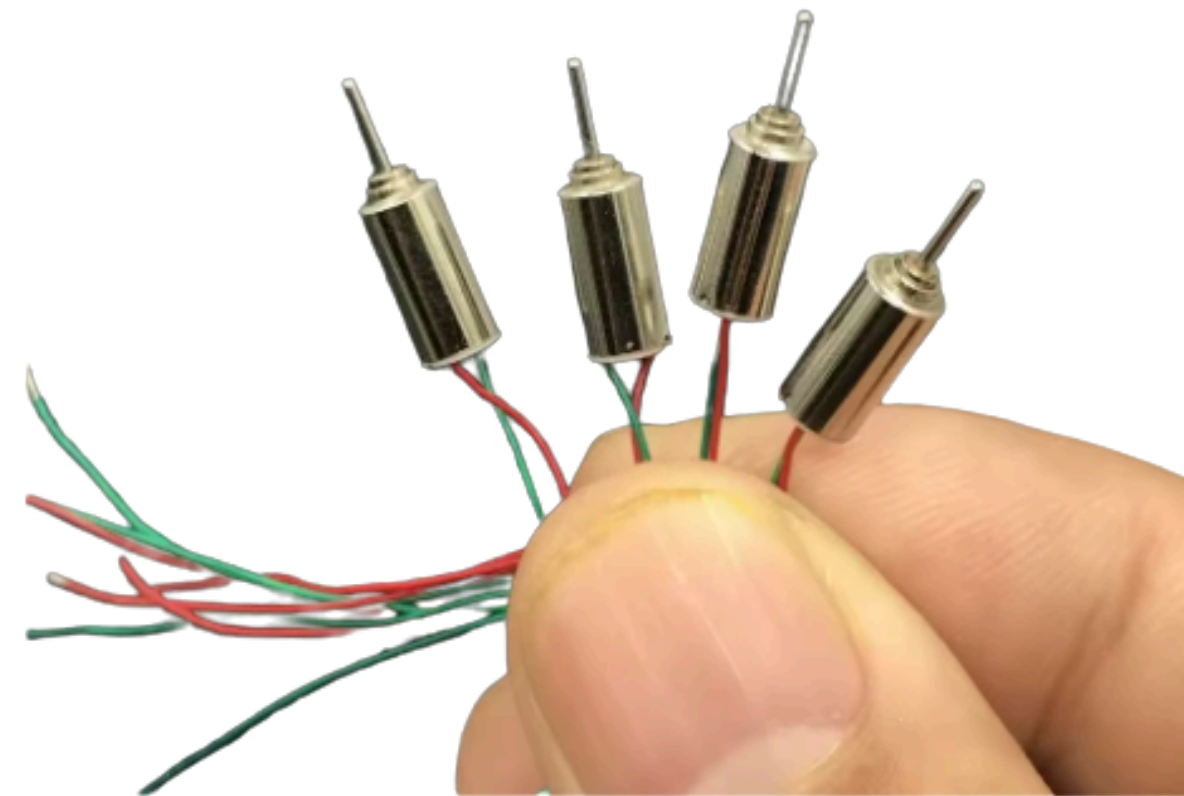
**Actuators that can be
used for the locking
mechanisms**

Compliant mechanism

3D printed, but it needs a motor to run the mechanism



Found these drone motors online to actuate the mechanism



https://de.aliexpress.com/item/1005004940678356.html?dp=CjwKCAIAwqHIBhAEEiwAx9cTeZMDoEP2y5Uzm6k4wJSD0Xw3dovcPUp0JbXl4V7xg94lUzkWd4aBhoCzBsQAvD_BwE&isd=y&aff_fsk=_Ewnfkmp&src=shoparizeDE&aff_platform=aff_feeds&aff_short_key=_Ewnfkmp&pdp_npi=4%40dis%21EUR%212.59%212.59%21%21%21%21%40%211200031093605854%21aff%21%21%21

Electromagnet



This is the
electromagnet used
for this mechanism

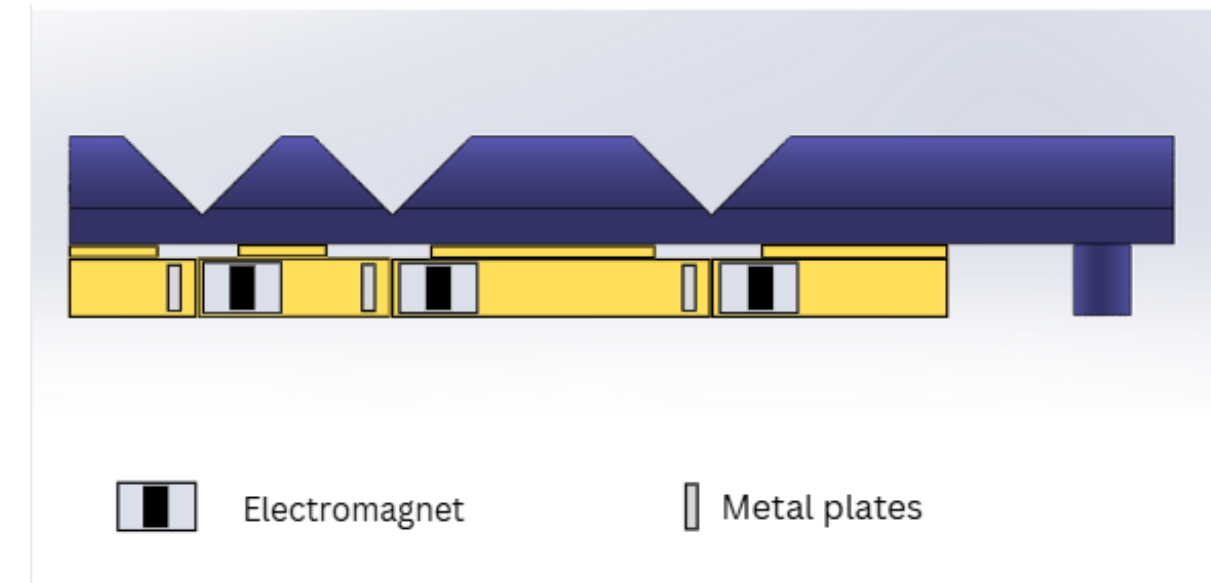


**20pcs Plates Iron Sheet Galvanized
3.93"x0.39", Thickness 0.01" 100x10x0.3mm Anti-Rust...**

20pcs Plates Iron Sheet Galvanized 3.93"x0.39", Thickness 0.01"
100x10x0.3mm Anti-Rust Folding DIY Handmade Small Thin Metal Foil
Strip: Amazon.com: Industrial & Scientific

 Amazon.com

These are the
metal sheets
used for the
mechanism



This is a sample
of what the
finger might
look like with
this mechanism

Linear actuators

I have 2 options, let's try them both to get a conclusion



https://de.aliexpress.com/item/1005005239069318.html?dp=CjwKCAiAwqHIBhAEEiwAx9cTeRKVZCpX3wcwoWHVgigAFd-InJXc2Jf0kMbZ_7X_kYRUL91zEmsAvxoC8_IQAvD_BwE&isdl=y&aff_fsk=_Ewnfkmp&src=shoparizeDE&aff_platform=aff_feeds&aff_short_key=_Ewnfkmp&pdp_npi=4%40dis%21EUR%210.95%210.95%21%21%21%21%21%40%2112000050605478027%21aff%21%21%21



Motor-C

https://de.aliexpress.com/item/1005010056837771.html?dp=CjwKCAiAwqHIBhAEEiwAx9cTeSQ7QspAGhEjs8-x77c6uvdw-UCmX3Or3V8Lakl2jt8j08IAmD8XEhoC6IEQAvD_BwE&isdl=y&aff_fsk=_Ewnfkmp&src=shoparizeDE&aff_platform=aff_feeds&aff_short_key=_Ewnfkmp&pdp_npi=4%40dis%21EUR%213.99%213.79%21%21%21%21%21%40%2112000050976404245%21aff%21%21%21

Control

To move the finger, we can use 2 servo motors.



For now, we can use an Arduino to control the finger.

For each motor, we need to have a motor controller

Finally, we need to design a PCB for the whole hand for around 25 -23 actuators

Power consumption

Electromagnet

2 W per magnet

$2 \times 3 = 6$ Watt/finger

Everything is in the 3V range. I will have to calculate
for other actuators and update this page.

List of actuators to buy

- [Drone motors - 4](#)
- [Electromagnets - 4](#)
- [Metal Sheets](#)
- [Linear actuator - 4](#)
- [Linear actuator motor c - 4](#)
- Arduino
- Motor controller - 3 or 15
- Servo motors - 2

The number
is the
number of
things to buy